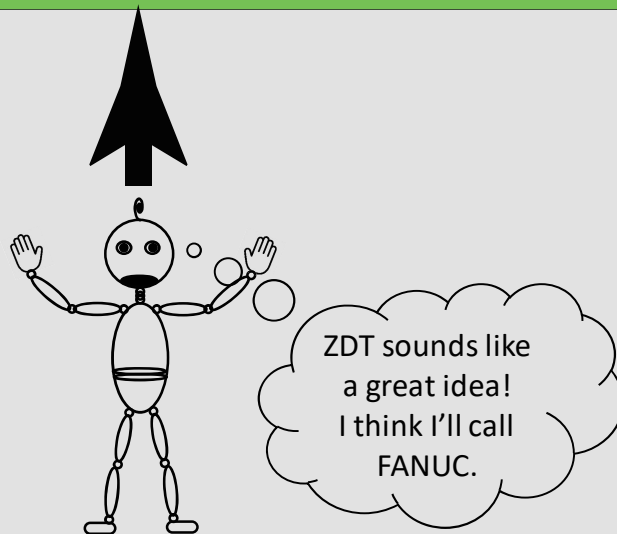
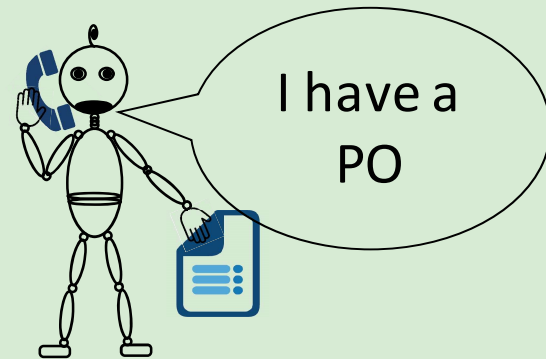


1. Portal Setup

Once you purchase ZDT, FANUC will create the portal through which you will monitor ZDT activities.

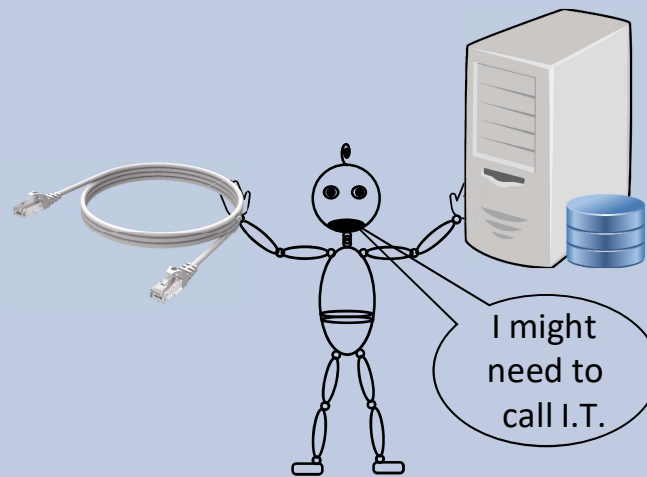
Once the portal is created, you can log in and set up your portal for your location.



2. Data Collector Configuration

The Data Collector collects and sends out data regarding robot health, maintenance, and connection status.

Connect the Data Collector to ensure a safe, reliable, and secure connection from the robot to FANUC.



4. ZDT Operations

At this point, ZDT is active. Once you log in to the portal and assign any unassigned robots for monitoring, you can use the portal for monitoring, and take action on notification alert emails.

More information on using the ZDT web portal for these functions can be found in its online help system.



3. Robot Setup

Load any necessary robot options and enable ZDT functions. Program any necessary self-tests for ZDT reducer monitoring.



Additional information regarding robot setup can be found in the *ZDT User's Guide* manual.

A. Log in to the Portal

- When FANUC cRc receives the P.O. and ZDT questionnaire, they will create your ZDT Web Portal and send you an invitation similar to the one shown below.



- When you get the invitation, click on **FANUC Robotics : Register** and fill out the registration form.

After agreeing to the listed policies, press the **Register** button.

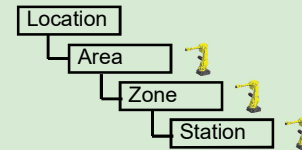
- Once the registration is complete, you will be able to log into your ZDT Web Portal.

B. Create Your First Area

Note: Initially, your ZDT portal will be empty.

You must start to do some initial configuration, such as create your site's areas, zones, and stations.

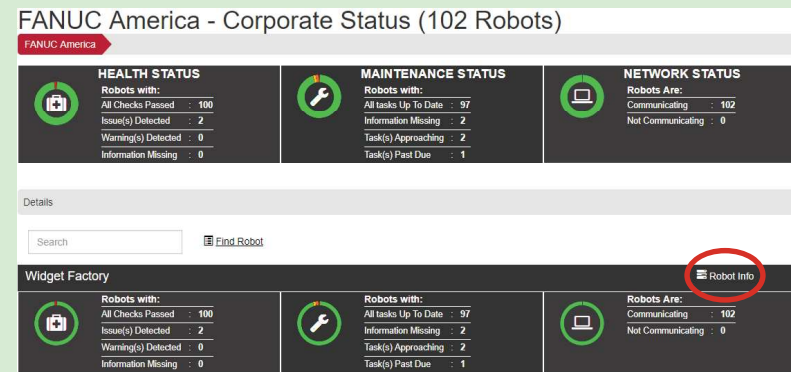
The ZDT portal organizes robots in a hierarchy.



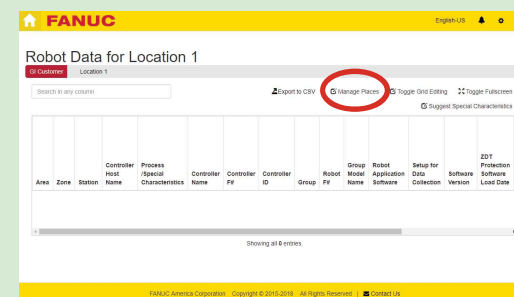
The Location is the highest level of organization in ZDT. When FANUC cRc created your web portal, they added your location for you. However, Locations cannot have robots directly assigned to them. Robots must be assigned to an Area, at the minimum.

Note: ZDT analysis does not begin until the robot(s) have been assigned to their area within the facility.

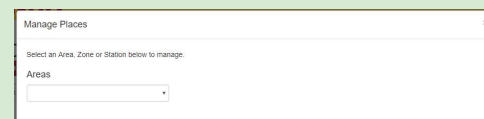
- Click on the Robot Info link corresponding to your location.



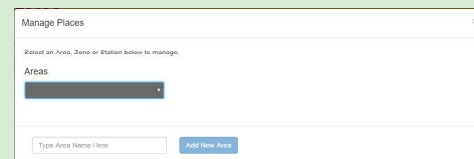
- Click on Manage Places in the Robot Data grid.



- Click on the Areas dropdown.



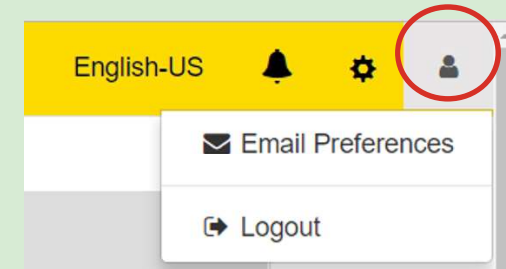
- Type the name of your new area, then press the **Add New Area** button.



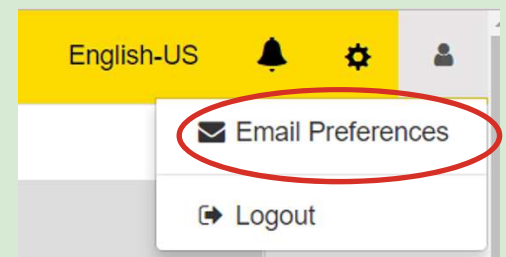
C. Set Email Preferences

The initial user in the portal is the company administrator. To set the email preferences for this user perform the following steps.

- Click the User icon in the upper right.



- Select Email Preferences.



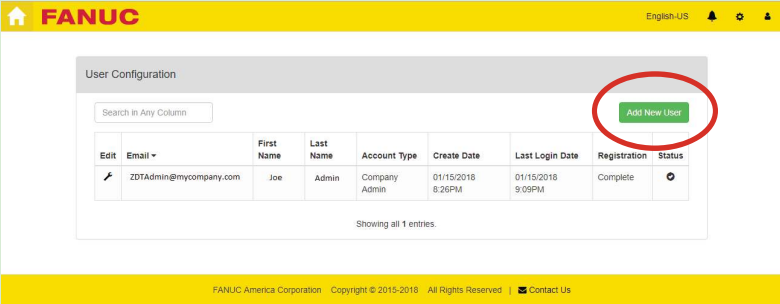
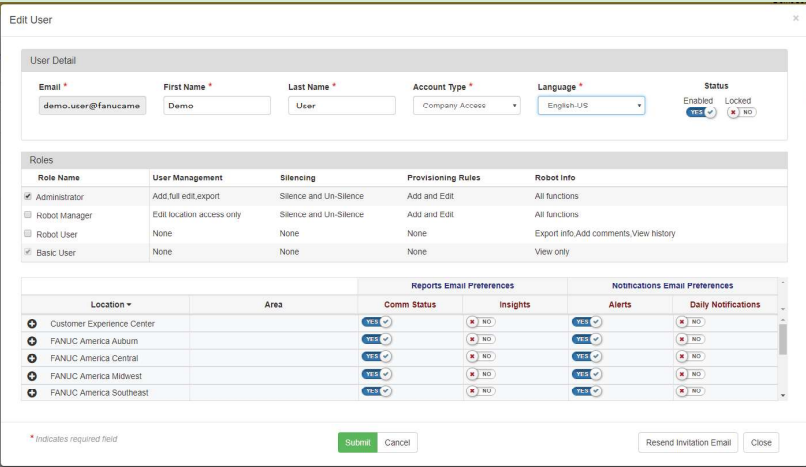
- For each location and area at your site, set the checkbox for the reports and notifications that the company administrator should receive.

These reports and notifications are defined as follows:

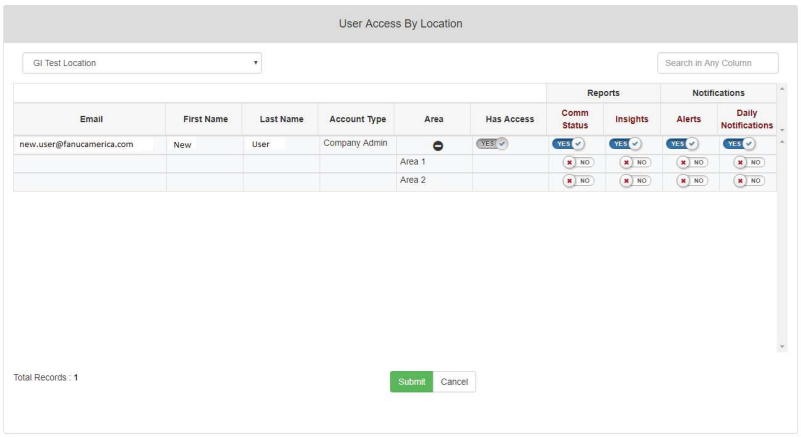
- Comm Status:** Daily email used to track which robots at a location have been sending messages to the ZDT Data Center.
- Insights:** Monthly email summarizing how robots at a location are functioning and where improvements can be made.
- Alerts:** Automatic emails sent by analytics when issues are identified, that require immediate attention.
- Daily Notifications:** Daily email summarizing how many alerts, warnings, and upcoming maintenance tasks have been detected at a location that day.

- Press **Submit** to save the changes.

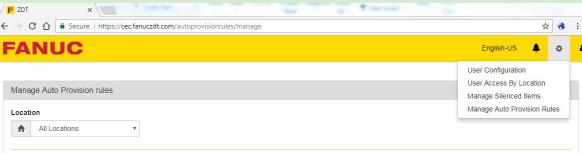
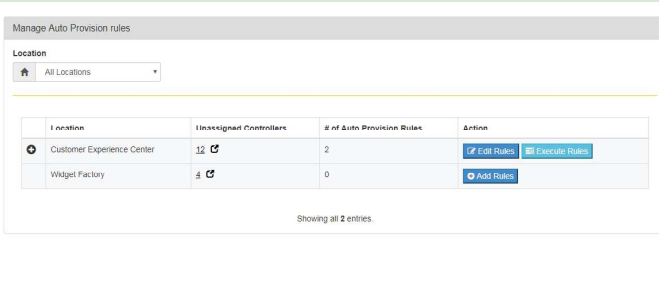
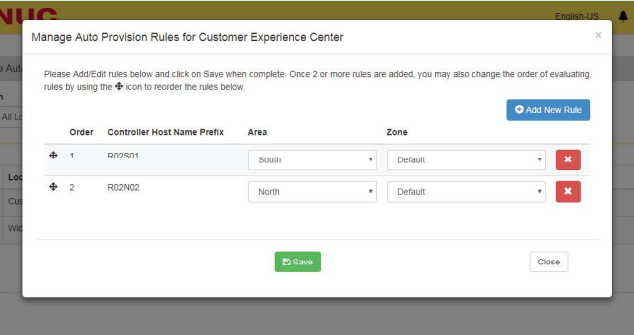
D. Add Other Users (Optional)

- Note:** The company admin user can add additional users and allow them access to the system.
- Click the Gear icon in the upper right.
 - Choose **User Configuration**.
 - Click **Add New User**.
- 
- Fill in the Email (must be a company email address), First Name, and Last Name fields.
- 
- Set an appropriate Account Type:
 - Company Access:** User is able to perform their assigned roles in all areas of all locations within the company.
 - Location Access:** User is able to perform their assigned roles at all areas within specific locations in the company, as assigned by an administrator.
 - Area Access:** User is able to perform their assigned roles only within specific areas at specific locations in the company, as assigned by an administrator.
 - Select the User's Role:
 - Administrator:** Can perform all user management functions, can silence/un-silence warnings, can add/edit auto provisioning rules, can perform all robot editing and management functions.
 - Robot Manager:** Can only grant/deny users access to locations, can silence/un-silence warnings, can add/edit auto provisioning rules, can perform all robot editing and management functions.
 - Robot User:** Cannot perform any user management functions, cannot silence/un-silence warnings, cannot do anything with auto provisioning rules, can only add and view robot comments.
 - Basic User:** Cannot perform any user management functions, cannot silence/un-silence warnings, cannot do anything with auto provisioning rules, can only view robot data.
 - Set the new user's default language.
 - Set the user's email notification preferences.
 - Press Submit.

E. Configure User Access (Optional)

- Note:** The company admin user can assign each user's access to different levels of the company hierarchy.
- Click the Gear icon in the upper right.
 - Choose **User Access by Location**.
- 
- Find the user to be configured in the list.
 - Use the **Has Access** checkbox to allow or deny the user access to different hierarchy levels at a location.
 - Use the **Reports** and **Notifications** checkboxes to send or not send the corresponding reports and notifications to the user.
 - Press Submit.

F. Rules for New Robot Connections (Optional)

- To have your newly connected robots automatically assigned to their area of the facility, you can set up auto provision rules.
- Click the Gear icon in the upper right.
 - Choose **Manage Auto Provision Rules** using the Manage Auto Provision Rules UI in the web portal.
- 
- From the dropdown menu, select your location and click on **Add Rules**.
- 
- Click on **Add New Rule**.
- 
- Enter your **Controller Host Name Prefix**.
 - Select the target area.
 - If you have a zone, select the target zone.
 - Press **Save**.

Portal setup is now complete.
Now refer to the Data Collector Startup Guide.

For more information
1-888-FANUC-US (1-888-326-8287)
zdtadmin@fanucamerica.com
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Requirements

Note: This Startup Guide assumes that your robot is not connected to a network.

The ZDT IR809 Data Collector requires the following:

	Requirement	Description
IR809	24V DC power source	
	ZDT provisioning file (.zxl)	(Required) ZDT provisioning file will be provided by FAC
	Certificate for HTTPS authentication	(Optional)
Robot	Robot Controller version	V8.30 or higher
	Robot ZDT software	Preinstalled on R-30iB Plus R882 on V8.30p29 or greater
	Number of robots	GI Data Collector currently supports up to 16 robots
Network	Network architecture	GI Data Collector must function as a network router
	Broadband network	
	Network switch	An unmanaged or managed switch for connecting robots, IR809, and PC
	Ethernet cables	
	Customer subnet configuration	Customer specify IR809 router subnet address scheme for connected robots and devices
ZDT portal account	Network information and access	DHCP service with DNS, NTP for 809/829 northbound Firewall rules for FANUC and Cisco cloud
	Administrator Corporate site structure Notification users	Administrator of the admin account with a list of corporate site data and users to be notified by ZDT
PC	Supported browser	(Required) Latest version of Chrome (Version 61.0.3163.100) or IE 11. Note If you receive a message that a log or other file cannot be downloaded when using IE 11, install the latest Microsoft security update.
	Supported resolution	(Required) Minimum 1024x768

The following TCP network ports must be open to allow the ZDT GI Data Collector to communicate with the robots and the cloud.

	Ports	Purpose
Robot Network	Robot Controller to IR809 switch	TCP 6062, 6063 Robots send ZDT messages to ports 6062 and 6063 of the data collector
	IR809 to Robot Controller switch	TCP 3002 The data collector sends and receives data from robots using port 3002
	IR809 to Robot Controller switch	TCP 21 Data sent and received via FTP
	PC and IR809 switch	TCP 7443 For accessing Data Collector web interface
Internet	IR809 to FANUC data center	TCP 443 The data collector sends secure messages to the cloud using URL fanuc01.cisco.com on port 443
	IR809 from/to Cisco Kinetic	UDP 500, 4500 TCP 9123, 9124 Bidirectional UDP 500, 4500 to us.ciscokinetic.io and TCP 9123, 9124 to us.ciscokinetic.io

Requirements

Network Information

Your system consists of robot(s) not connected to a plant or other network.

To make deployment of the ZDT system simpler, the default network configuration for the IR809 is as follows (**Note:** The GE1 port and the following network configuration will not be active until the IR809 has been provisioned):

- IR809 IP: 192.168.1.1
- IR809 Reserved Static IP Range: 192.168.1.2 - 192.168.1.128
- IR809 Subnet Mask: 255.255.255.0
- DNS Server IP: 192.0.2.254

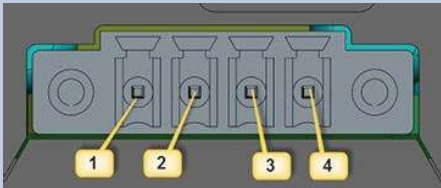
Note: Your IT department may only allow approved devices to communicate on the network. In this case, even if you wire the IR809 properly, it won't be able to communicate unless IT provides access. Please contact your IT department and make sure the connection for the IR809 meets the stated requirements.

A. Power Up and Connect the IR809 to the Network

1. Write down your IR809 serial number. This will be used at a later time.



2. Supply 24VDC power to the IR809 unit.
Note: The power supply may be customer-supplied, or purchased from FANUC.

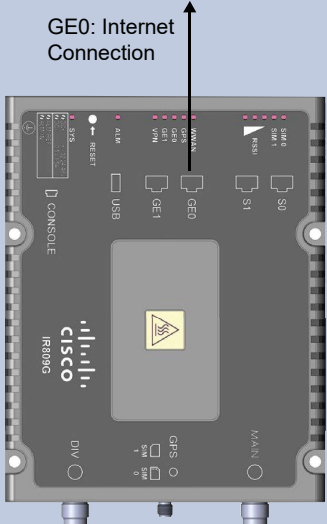


Pin 1: DC In +
Pin 2: DC In -
Pin 3: Not used.
Pin 4: Not used.

3. Integration into networks: Ethernet cable wiring. In a Gateway configuration, the IR809 requires two network configurations. One port connects to the Internet, and one port connects to the network that contains the robots.

- a. Establish the connection to the Internet
Wire the Internet connection to the GE0: port.

Note: The GE1 port will not be active until the IR809 has been provisioned.



A. Power Up and Connect the IR809 to the Network (continued)

Before proceeding to the next step, validate the LED Status below.

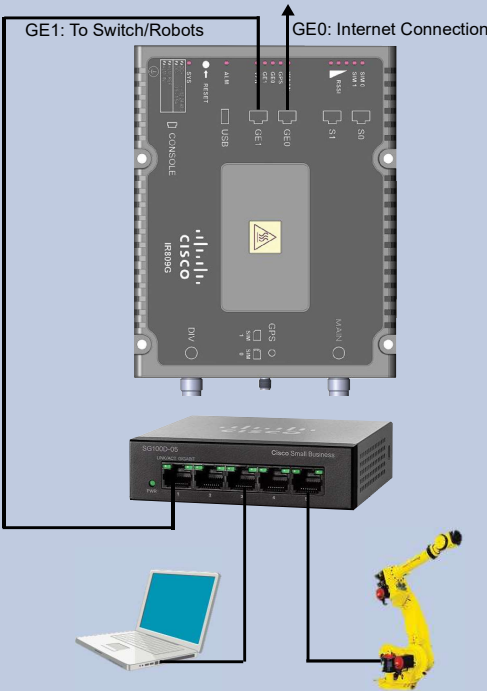
- Sys LED: Green
- GE0 LED: Green

If the LED's are not all Green after 10 minutes, this generally indicates an issue with system status or connectivity. Validate that all Port requirements are met. Contact FANUC cRc if additional assistance is required.

IR809 LEDs		
LED	Activity	Description
SYS	Power and System Status	Off — No power Green Steady on — Normal operation Green Flashing — Boot up phase or in ROM Monitor mode Amber Steady on — Power is OK but possible internal FPGA program failure
VPN_OK	VPN	Off — No VPN tunnel Steady Green — At least one VPN tunnel is up
GE0 (10/100/1000)WAN 0 GE0 (10/100/1000)WAN 1	Link Status	Off — No link Steady Green — Link is up Flashing — Transmitting and Receiving data

- b. Establish the robot network

Add an unmanaged switch and connect a laptop/PC, your robot port (CD38A or CD38B), and the IR809 port (GE1) to the switch. **Make note of the Robot Port to which you connected. You will need this information when communicating with FANUC cRc.**



B. Provision the IR809

Contact FANUC cRc (zdtsiteadmin@fanucamerica.com) to notify them that your device has been connected to the internet.

Provide:

- Serial number of the IR809
- Robot (port CD38A or CD38B) connected to the switch as noted in step A3b.

FANUC will:

- Register your product serial number to the ZDT System to allow configuration and communication operations.
- Prepare your software in the cloud and begin downloading automatically.
- Inform you when the software download is completed.
- Provide you with:
 - ZDT GI Data Collector web interface URL to enable you to access your data collector
 - .ZXL file which is used to configure the IR809 to send data to the ZDT Data Center
 - RCMCFG.XML file which is used to configure your robots to communicate with your IR809. This XML filename may have additional description text and will be used in the Robot Setup Startup Guide

Note: The GE1 port is now active and may be used.

C. Configuration

1. You will now use the laptop/PC connected in step A3b. If your laptop/PC has DHCP enabled (typical), the laptop/PC can obtain an IP/Subnet through the IR809 DHCP server automatically. If your laptop/PC does not have DHCP enabled, change the laptop/PC settings to use an unused unique IP address. Robot network settings will be covered in the *Robot Setup Startup Guide*.

2. Connect a laptop/PC to the IR809 GE1 port to view the web UI.

Note: It is required that a switch be used to support IR809, robots, and PC to communicate.

3. Make sure that the laptop/PC is configured to match network settings described in Requirements Network Information section.

Note: If the laptop/PC has multiple network interfaces (WiFi, Ethernet), disable all network interfaces except the connection to GE1: switch.

4. To access the Data Collector Web UI do the following:

- a. Using the laptop/PC connected in step A4, open the web browser (IE or Chrome) and enter the ZDT GI Data Collector web interface URL provided in section B. e.g. https://10.8.188.122:7443

Notes: The port number 7443 is required.
The protocol needs to be **https**, not http.

- b. Proceed to the Data Collector login page. IE and Chrome work slightly differently as outlined below:

• **If IE is the web browser**

A screen similar to the one below will appear. You will need to click on **More information**.

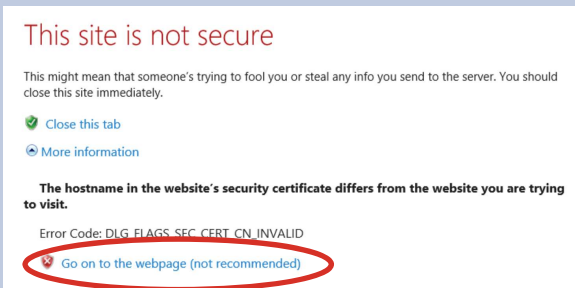
This site is not secure

This might mean that someone's trying to fool you or steal any info you send to the server. You should close this site immediately.

- Close this tab
- More information

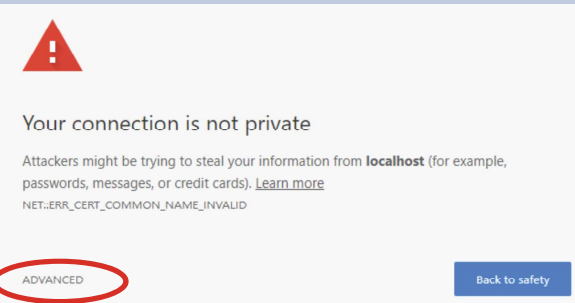
C. Configuration (continued)

The screen will expand to look similar to the following. Click **Go on to the web page (not recommended)**.

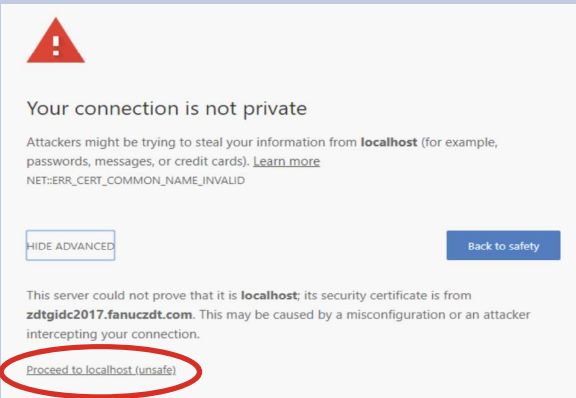


• **If Chrome is the web browser**

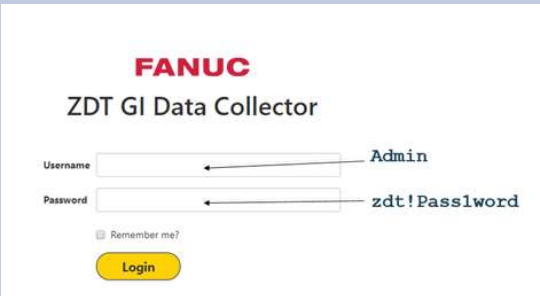
On the screen that appears, you will need to click on **Advanced**.



The screen will expand to look similar to the following. Click on **Proceed to <your IP address> (unsafe)**.



C. Whether you used IE or Chrome to access the Data Collector, as outlined in step C4b, you will be taken to the Data Collector login page as shown below. If you are able to view the Data Collector Web UI, it confirms that the Data Collector is running at the given IP address.

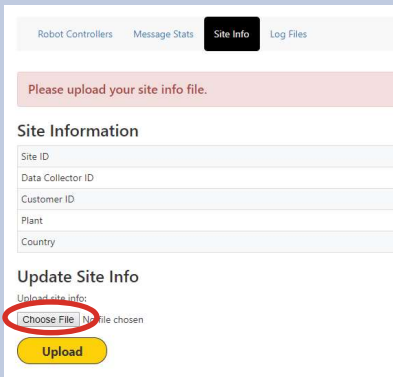


- d. Login in to the ZDT GI Data Collector with the following default username and password:
Username: admin
Password: zdt!Pass1word

Note: Since the system has a default password, you will be asked to change this default password. **FANUC recommends that you change your password.** Change the password and select **ZDT Data Collector** from the Side Menu or, if you choose not to change your password, simply select **ZDT Data Collector** from the Side Menu to proceed. You will be taken to the screen shown in step C5.

C. Configuration (continued)

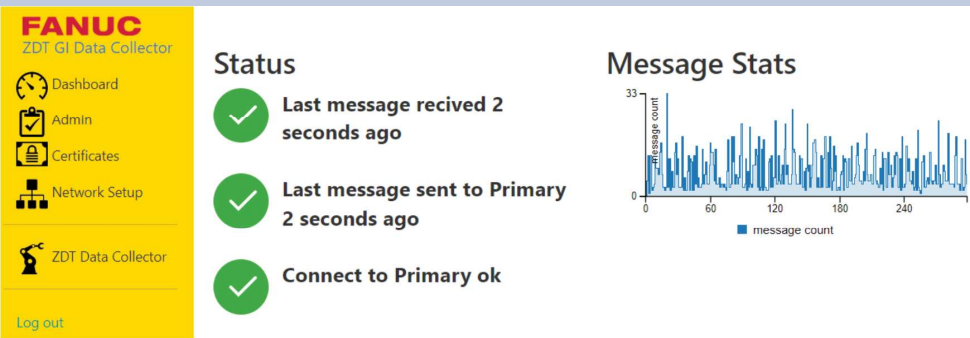
5. You will see a screen similar to the following. Press **Choose File** to select the ZXL file provided by FANUC.



6. Press **Upload** to install the site info file (ZXL file) provided by FANUC.

D. Verification

1. On the ZDT Data Collector Web UI, use the Data Collector Dashboard to check the status. Green indicates that the ZDT Data Collector is able to receive ZDT messages from the robot controller(s), and send them to the FANUC data center.



Status Descriptions

Message Received in Last x second(s)/minute(s): Indicates how much time has passed since the last message was received from the robots. Robots do not send messages to the Data Collector all of the time. When testing with one or more robots in idle, it is normal to see the status indicator turn green when you power on the robot(s) and then turn red after 15 minutes.

Connect to Primary ok: Red indicates that the Data Collector is unable to connect to the FANUC data center. Common issues are: The Data Collector has no network communication permission, there is a DNS failure, there are incorrect network settings, or communication is blocked by a corporate firewall.

You are ready to configure your robots.
Refer to the ZDT User's Guide manual.

For more information

Refer to FANUC America Corporation
ZDT Data Collector User's Guide

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